

Installation and Configuration of NS-3 with NetAnim on WSL for AODV and OLSR MANET Routing Protocol Visualization

Nilambar Elangbam

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1 Introduction

This document provides a complete step-by-step guide for setting up the NS-3 network simulator on Windows using Windows Subsystem for Linux (WSL), enabling graphical visualization through NetAnim, and running MANET routing protocols (AODV and OLSR) with animation support.

The objective is to create a fully functional simulation environment capable of:

- Running MANET routing protocol simulations.
- Generating animation XML trace files.
- Visualizing node movement and packet transmission.
- Observing routing behavior in AODV and OLSR.
- Performing MANET protocol analysis and comparison.

2 System Requirements

- Windows 11 (recommended for WSLg GUI support)
- WSL2
- Ubuntu 22.04 or Ubuntu 24.04
- Minimum 8GB RAM recommended
- Internet connection

3 Installing Windows Subsystem for Linux (WSL)

3.1 Enable WSL

Open PowerShell as Administrator and run:

```
wsl --install
```

Restart the system if prompted.

3.2 Install Ubuntu

After restart:

- Open Ubuntu from the Start Menu.
- Create Linux username and password.

4 Updating Ubuntu

Inside WSL terminal:

```
sudo apt update  
sudo apt upgrade -y
```

5 Installing Required Dependencies

```
sudo apt install -y build-essential \  
cmake git gcc g++ python3 python3-pip \  
libsqlite3-dev libboost-all-dev \  
qtbase5-dev qt5-qmake
```

6 Installing NS-3

6.1 Clone Repository

```
cd ~  
git clone https://gitlab.com/nsnam/ns-3-dev.git  
cd ns-3-dev
```

6.2 Configure NS-3

```
./ns3 configure --enable-examples
```

6.3 Build NS-3

```
./ns3 build
```

6.4 Verify Installation

```
./ns3 show targets
```

“latex

7 Enabling Animation in DSDV Simulation

7.1 Open DSDV Example

```
nano ~/ns-3-dev/src/dsdv/examples/dsdv-manet.cc
```

7.2 Add NetAnim Header

Add the following header at the top:

```
#include "ns3/netanim-module.h"
```

7.3 Add Animation Code

Before:

```
Simulator::Run();
```

Add:

```
AnimationInterface anim("dsdv-animation.xml");  
anim.EnablePacketMetadata(true);
```

7.4 Modify CMake File

Open:

```
nano ~/ns-3-dev/src/dsdv/examples/CMakeLists.txt
```

Modify:

```
LIBRARIES_TO_LINK  
    ${libdsdv}  
    ${libnetanim}
```

7.5 Rebuild NS-3

```
cd ~/ns-3-dev  
./ns3 clean  
./ns3 configure --enable-examples  
./ns3 build
```

7.6 Run DSDV Simulation

```
./ns3 run dsdv-manet
```

Generated output file:

```
/home/username/ns-3-dev/dsdv-animation.xml
```

8 Enabling Animation in DSR Simulation

8.1 Open DSR Example

```
nano ~/ns-3-dev/src/dsr/examples/dsr-example.cc
```

8.2 Add NetAnim Header

Add the following header at the top:

```
#include "ns3/netanim-module.h"
```

8.3 Add Animation Code

Before:

```
Simulator::Run();
```

Add:

```
AnimationInterface anim("dsr-animation.xml");  
anim.EnablePacketMetadata(true);
```

8.4 Modify CMake File

Open:

```
nano ~/ns-3-dev/src/dsr/examples/CMakeLists.txt
```

Modify:

```
LIBRARIES_TO_LINK  
    ${libdsr}  
    ${libnetanim}
```

8.5 Rebuild NS-3

```
cd ~/ns-3-dev  
./ns3 clean  
./ns3 configure --enable-examples  
./ns3 build
```

8.6 Run DSR Simulation

```
./ns3 run dsr-example
```

Generated output file:

```
/home/username/ns-3-dev/dsr-animation.xml
```

““

9 Installing NetAnim

9.1 Clone NetAnim Repository

```
cd ~  
git clone https://gitlab.com/nsnam/netanim.git  
cd netanim
```

9.2 Build NetAnim

```
mkdir build
cd build
cmake ..
make -j$(nproc)
```

9.3 Run NetAnim

```
./netanim
```

10 Enabling Animation in AODV Simulation

10.1 Open AODV Example

```
nano ~/ns-3-dev/src/aodv/examples/aodv.cc
```

10.2 Add NetAnim Header

Add the following header at the top:

```
#include "ns3/netanim-module.h"
```

10.3 Add Animation Code

Before:

```
Simulator::Run();
```

Add:

```
AnimationInterface anim("aodv-animation.xml");
anim.EnablePacketMetadata(true);
```

10.4 Modify CMake File

Open:

```
nano ~/ns-3-dev/src/aodv/examples/CMakeLists.txt
```

Modify:

```
LIBRARIES_TO_LINK
    ${libaodv}
    ${libnetanim}
```

10.5 Rebuild NS-3

```
cd ~/ns-3-dev
./ns3 clean
./ns3 configure --enable-examples
./ns3 build
```

10.6 Run AODV Simulation

```
./ns3 run aodv
```

Generated output file:

```
/home/username/ns-3-dev/aodv-animation.xml
```

11 Enabling Animation in OLSR Simulation

11.1 Open OLSR Example

```
nano ~/ns-3-dev/src/olsr/examples/olsr-hna.cc
```

11.2 Add NetAnim Header

Add:

```
#include "ns3/netanim-module.h"
```

11.3 Add Animation Code

Before:

```
Simulator::Run();
```

Insert:

```
AnimationInterface anim("olsr-animation.xml");
anim.EnablePacketMetadata(true);
```

11.4 Modify CMake File

Open:

```
nano ~/ns-3-dev/src/olsr/examples/CMakeLists.txt
```

Add:

```
${libnetanim}
```

Ensure:

```
LIBRARIES_TO_LINK
    ${libolsr}
    ${libnetanim}
```


11.5 Rebuild NS-3

```
cd ~/ns-3-dev  
./ns3 build
```

11.6 Run OLSR Simulation

```
./ns3 run olsr-hna
```

Generated output file:

```
/home/username/ns-3-dev/olsr-animation.xml
```

12 Visualizing Simulation in NetAnim

Launch NetAnim:

```
~/netanim/build/netanim
```

Open:

- aodv-animation.xml
- olsr-animation.xml

Press Play to visualize:

- Node movement
- Packet transmission
- Route discovery
- MANET topology changes
- Routing behavior

13 Common Errors and Solutions

13.1 Undefined Reference to AnimationInterface

Cause:

- NetAnim library not linked.

Solution:

- Add `$libnetanim` inside `LIBRARIES_TO_LINK`.

13.2 Qt Runtime Warning

`QStandardPaths: wrong permissions on runtime directory`

This warning is harmless in WSL.

13.3 Simulation Does Not Generate XML File

Cause:

- AnimationInterface not added correctly.

Solution:

- Verify animation code is placed before `Simulator::Run()`.

13.4 Build Errors

Fix:

```
./ns3 clean
./ns3 configure --enable-examples
./ns3 build
```

14 Conclusion

The NS-3 network simulator was successfully configured on WSL with graphical visualization support using NetAnim. The AODV and OLSR MANET routing protocols were executed and visualized successfully.

This environment enables:

- MANET routing experimentation
- Protocol comparison
- Network topology visualization
- Packet-level routing analysis
- Academic and research-oriented simulation studies

The setup can be further extended for:

- DSR and DSDV simulations
- Performance metric analysis
- Throughput and delay measurements
- Wireless sensor network simulations
- VANET and IoT research